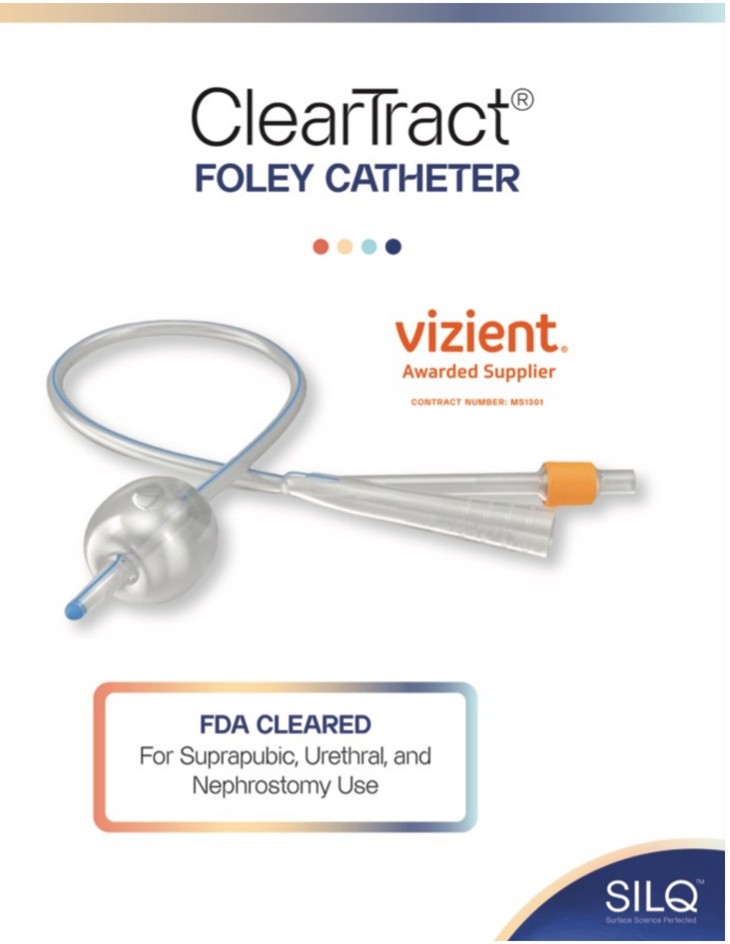
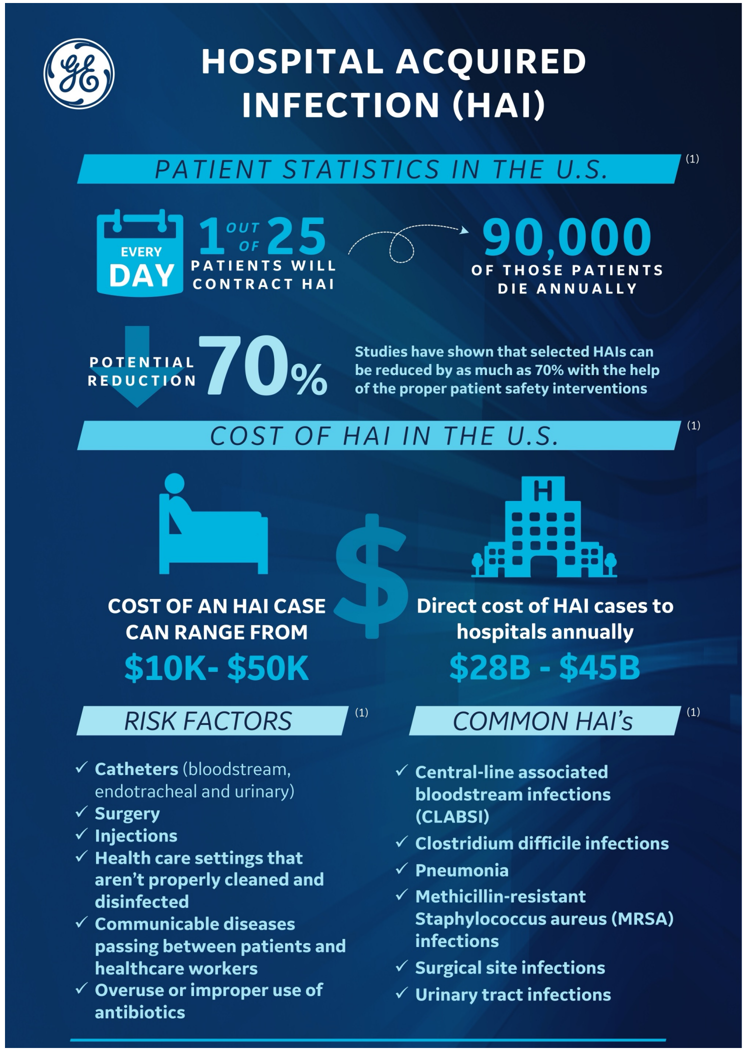
SILQ Surface Science Perfected.



ClearTract® Catheter

Brian McVerry, Ph.D.
Chief Technology Officer

SILQ Surface Science Perfected.



By **JESSICA FIRGER** / **CBS NEWS** / *March 26, 2014, 5:39 PM*

In U.S., hospital-acquired infections run rampant

SEP 03 | MORE ON QUALITY AND SAFETY

COVID-19 increased the number of healthcare-acquired infections

Central line infections increased by at least 46%, compared to 2019.

“…the overall annual direct medical costs of HAI to U.S. hospitals ranges from $28.4 to $33.8 billion…”

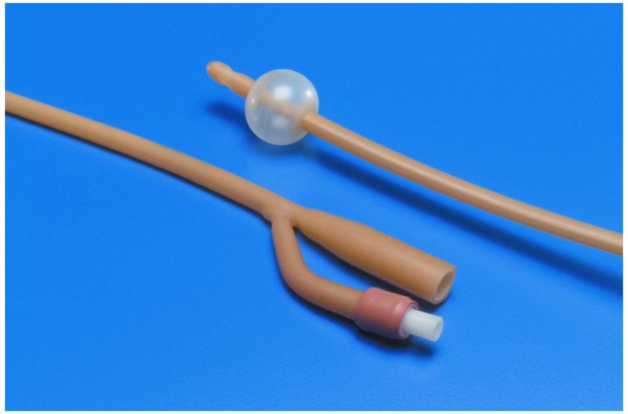
The Direct Medical Costs of Healthcare-Associated Infections in U.S. Hospitals and the Benefits of Prevention, 2009

-R. Douglas Scott II, ***Economist***

Catheterization is associated with high complication rates

Catheter-associated complications:

- Catheter-associated urinary tract infection (CAUTI)
- Blockage
- Hematuria
- Leakage
- Urethral stricture or erosion



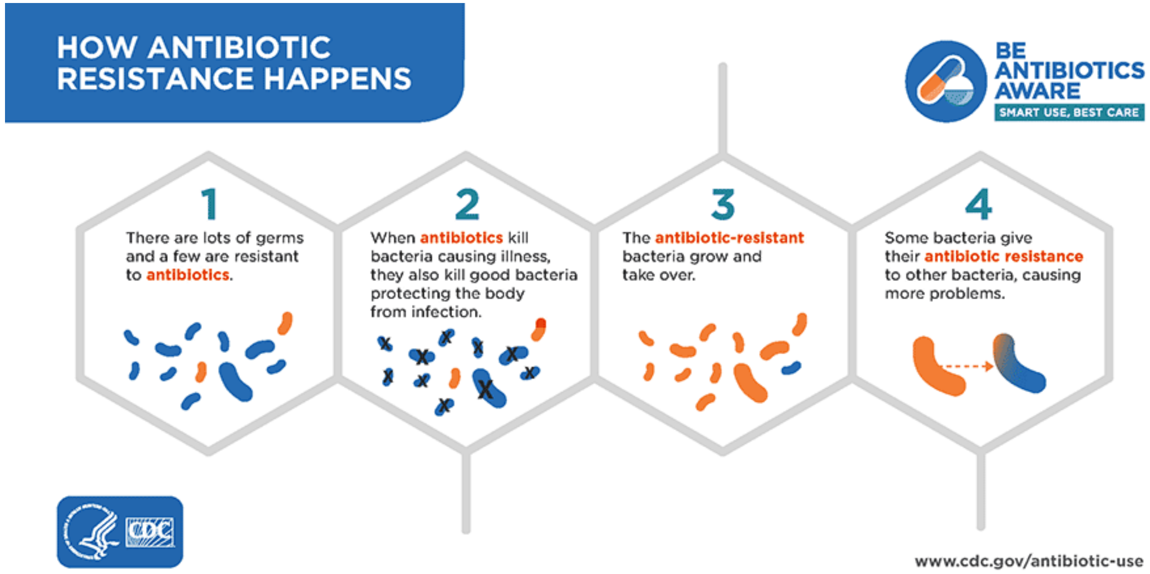
"CAUTIs are the most common nosocomial infections, and account for 1 million cases per year in the United States."

Catheter-Associated Urinary Tract Infections: Current Challenges and Future Prospects. **Res. Rep. Urol**. 2022, 14:109-133.

"Moreover, long-term catheterization and catheter use in patients with spinal cord injury results in even greater illness, with more than 30% of patients having several complications."

Determining the Noninfectious Complications of Indwelling Urethral Catheters: a systematic review and meta-analysis. **Ann. Intern. Med.** 2013, 159(6):401-10.

Conventional attempts to reduce catheter complications may do more harm than good



Catheters infused with antibiotics or antimicrobial silver aim to kill bacteria, proliferating antibiotics resistant microbes.

Several studies have demonstrated that impregnation of antibiotics or silver may reduce bacteriuria from urine culture short-term, but have not noted any clinic benefit or improvement compared to standard catheters.

A prospective trial of a novel, silicone-based, silver- coated foley catheter for the prevention of nosocomial urinary tract infections. **Infect Control Hosp. Epidemiol.** 2006, 27(1):38-43.
Antimicrobial catheters for reduction of symptomatic urinary tract infection in adults requiring short-term catheterisation in hospital: a multicentre randomised controlled trial. **Lancet**. 2012, 380(9857):1927-35.
Efficacy of Antimicrobial Catheters for Prevention of Catheter-Associated Urinary Tract Infection in Acute Cerebral Infarction. **J. Epidemiol.** 2018, 28(1):54-58.
Silver or nitrofurazone impregnation of urinary catheters has a minimal effect on uropathogen adherence. **J. Urol**. 2010, 184(6):2565-2571.

The Society for Healthcare Epidemiology of America advises against the routine use of antimicrobial impregnated catheters.

Strategies to prevent catheter-associated urinary tract infections in acute care hospitals: 2014 update. **Infect Control Hosp. Epidemiol**. 2014, 35 Suppl 2: S32-47.

The goal is to reduce Biofilm, not bacteriuria

1. Bacterial cells attach to a surface.
2. The cells secrete substances that strengthen their adhesion to the surface and to each other.
3. The cells proliferate to form biofilms, facilitating the sharing of genetic information including antibiotic resistance genes
4. The biofilm continues to grow and causes infection. The thick film blocks the penetration of antibiotics or silver-ions into the film.
5. Left unchecked, the biofilm eventually releases planktonic cells to colonize other parts of the catheter surface and repeat the cycle

CAUTI | CATHETER ASSOCIATED UTI

Diagnostic stewardship for urinary tract infection: A snapshot of the expert guidance

ABSTRACT

The urine culture, the cornerstone for laboratory diagnosis of urinary tract infection (UTI), is associated with a high frequency of false-positive and false-negative results, and its diagnostic threshold is debated. Urine culture takes days to result, and antibiotics are often initiated while awaiting final culture readings. Further, asymptomatic bacteriuria—the presence of bacteria in urine in the absence of UTI symptoms—generally does not warrant treatment.

URINARY TRACT INFECTIONS (UTIS) are among the most common human infections. But the urine culture, the cornerstone for laboratory diagnosis of UTI, is imperfect. It has a high frequency of false-positive and false-negative results, and its diagnostic threshold is debated. Also, urine culture takes 2 to 3 days to result, and antibiotics are often initiated empirically while awaiting final culture readings. Further, asymptomatic bacteri-

"Our results question the clinical value of commercially available antimicrobial urinary catheters."

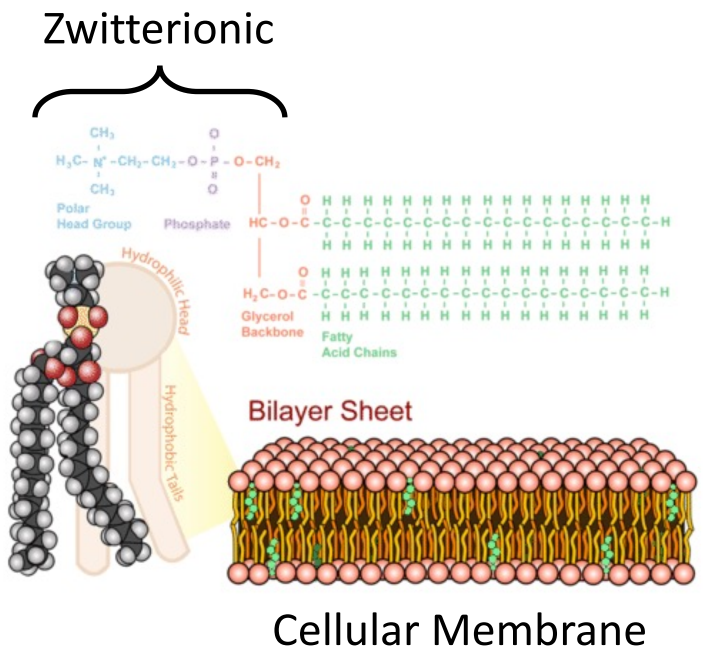
Silver or nitrofurazone impregnation of urinary catheters has a minimal effect on uropathogen adherence. **J Urol.** 2010 Dec;184(6):2565-71

"…no material has been found to date which eliminates bacterial colonization and biofilm formation."

Role of noble metal-coated catheters for short-term urinary catheterization of adults: a meta-analysis. **PLOS ONE** 2020 15(6): e0233215.

The power of Zwitterion chemistry targets the source: Biofilm

Zwitterion chemistry surrounds every cell in our body. For the past 20+ years, biomaterial researchers have studied the material's unique ability to resist biofilm formation, microbial deposition, and organic adhesion. However, applying zwitterion chemistry to the surfaces of devices is a complicated process with highly specialized equipment, limiting zwitterion technology to the research environment.



ADVANCED MATERIALS

Research Article

A Readily Scalable, Clinically Demonstrated, Antibiofouling Zwitterionic Surface Treatment for Implantable Medical Devices

Brian McVerry, Alexandra Polasko, Ethan Rao, Reihaneh Haghniaz, Dayong Chen, Na He, Pia Ramos, Joel Hayashi, Paige Curson, Chueh-Yu Wu, Praveen Bandaru, Mackenzie Anderson ... See all authors

First published: 22 March 2022 | https://doi.org/10.1002/adma.202200254 | Citations: 11

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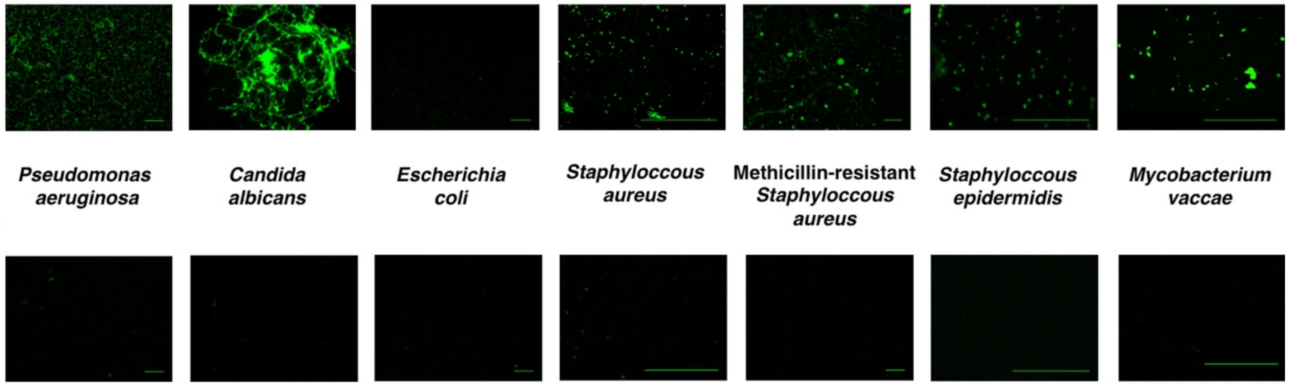
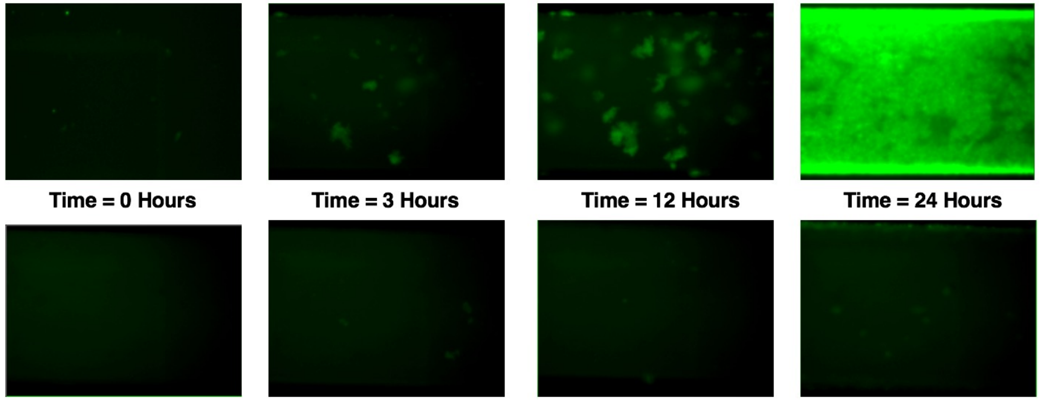
Researchers at UCLA have developed a scalable technique to apply zwitterion technology to the surface of medical devices, using common medical device manufacturing processes. It was these scientists who launched Silq to resists biofilm and complications from medical devices. Silq's first product is the ClearTract® indwelling Foley catheter.

Untreated Silicone Biofilm Adhesion



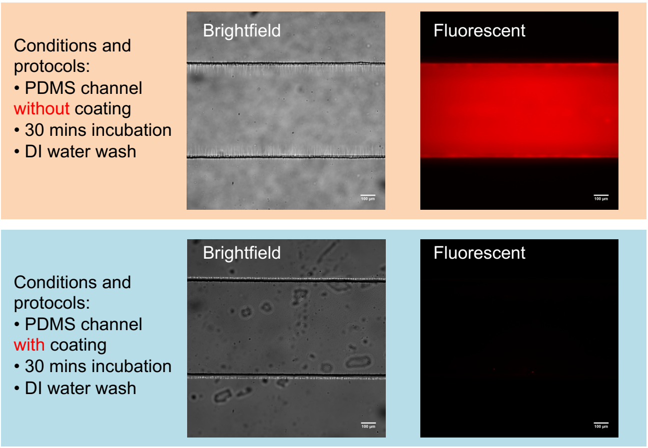
Zwitterion-treated Silicone Biofilm Adhesion

Untreated Silicone Channel

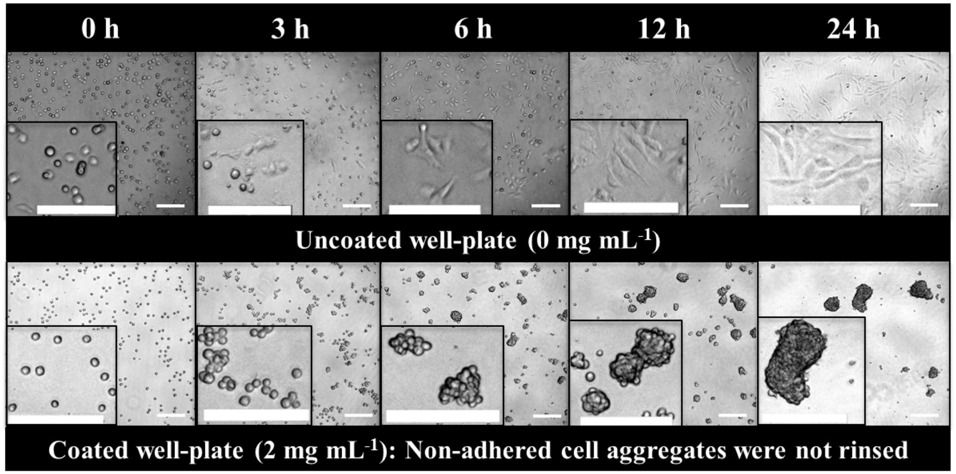


Zwitterion-treated Silicone Channel

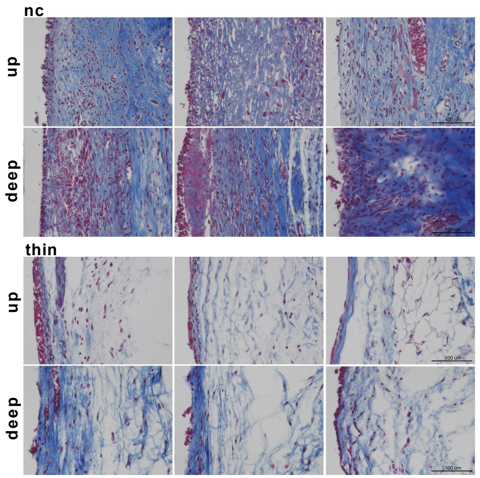
Fibrinogen Adhesion



Fibroblast Adhesion



Mice Tissue Histology

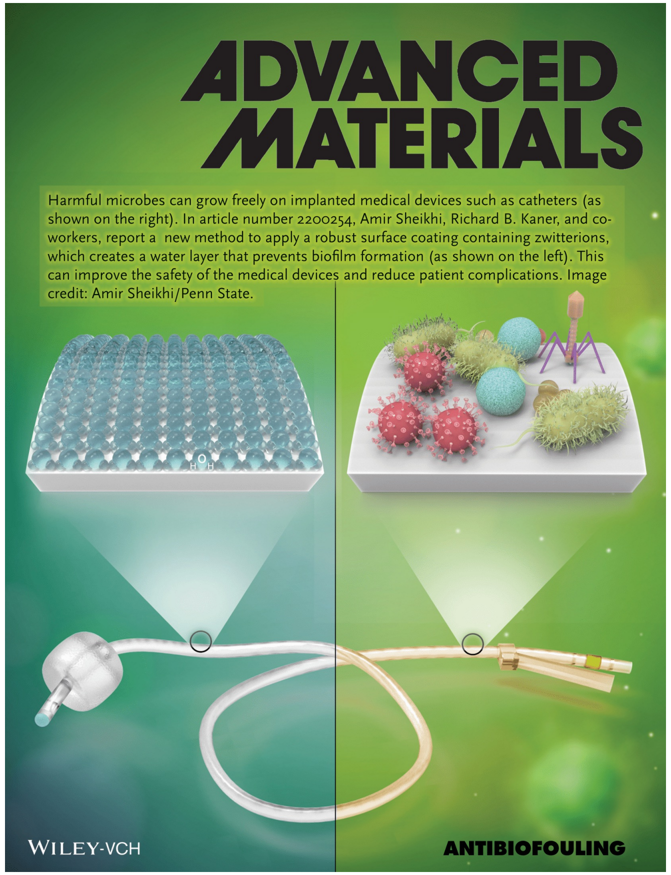


Surface Science Perfected.

ClearTract® Catheters

ClearTract® Features:

- Inner lumen and outer surface is treated with a permanent zwitterionic surface treatment
- Standard Foley catheter dimensions, specifications, and indications for use
- Made from 100% medical grade silicone
- Meets the FDA's newest (2020) guidelines for safety and biocompatibility for Foley catheters, including suprapubic
- Non-pyrogenic
- **Manufactured in the United States**



Front Cover of *Advanced Materials*
Published in March 2022

SILQ Surface Science Perfected.

ClearTract® Clinical Evidence

I. Non-randomized Control Study

Advanced Materials (Impact Factor: 32):

Study describes the first-time zwitterion-treated catheters were implanted in subjects requiring chronic catheterization

- Subjects with severe and frequent complications trialed ClearTract® at National Rehabilitation Center
- 62% of subjects described their urinary tract condition post implantation as "very much better" or "much better"
- 76% of subjects elected to remain on ClearTract® after exiting the study

Explanted Catheter Type	Total Number of Patients	Biofilm Quantity (ng ATP)
Bardex IC	35	337.24
Rusch Gold	39	289.50
Silq ClearTract	38	82.42

***Completed results for NCT0484122**

II. Randomized Clinical Trial

(NCT0484122):

Prospective, Multi-Center, Randomized Study to Assess the Ability of ClearTract® Urinary Catheter to Reduce Biofilm Formation

- Conducted at 8 Sites across the United States
- Quantifies biofilm on catheter surfaces upon explant in long-term catheterized subjects
- Compared ClearTract® against two of the most popular silicone-coated latex and silver-hydrogel "infection control" catheters
- ClearTract® demonstrated >70% reduction in biofilm compares to standard latex and Bardex IC

III. Randomized Clinical Trial

(NCT05931887):

Evaluation of Catheters in Subjects With Chronic Urinary Retention to assess the ability of ClearTract® to reduce catheter associated complications

- Randomized crossover study comparing complication rates vs. standard catheters
- Economic outcomes will be quantified based on reduction in complication rates
- Enrollment active/pending at the following sites:
  - Cleveland Clinic (Dr. Howard Goldman)
  - Rancho Los Amigos National Rehabilitation Center (Dr. Evgeniy Kreydin)
  - Santa Clara Valley Medical Center (Dr. Chris Elliott)
  - University of Virginia (Dr. Jacquelyn Zillioux)
  - Loma Linda University (Dr. Forrest Jellison)

Explanted Catheter Type	Catheters explanted	Treated UTIs	Ave. Mass Encrustation (mg)
Standard of Care	181	39	141.1
Silq ClearTract	184	9	47.9

***Current results for NCT05931887 from 79 subjects enrolled as of early February, 2025**
***34 out of 36 subjects that had a preference chose ClearTract® over standard of care (7 had no preference)**

SILQ Surface Science Perfected.

ClearTract® Clinical Evidence

Sometime in the last year the University of Michigan Hospital Urology Clinic received the SILQ sp catheter and part of samples to try. The first time this SILQ catheter was used we saw an immediate improvement in there being no clogging or sediment buildup in the catheter. Right from the first replacement we have been able to 6 weeks between replacements (the recommended interval) and there was still no sediment buildup in the catheter.

Needless to say, this has been a life-changing event for not only Nathan, but also his group home and us, his parents who take him to his appointments!

This has been like a miracle for us *and are unbelievably grateful for Brian McVerry and the SILQ Corporation for this incredible breakthrough and hope that this technology will soon be available widespread to any and all who need it.*

-Stephen Newhouse, 2023

When I was approached with the option to try the Silq catheter I was excited to see if it would help. From the time I started using Silq everything changed, I no longer had to flush with anything whatsoever. I went over 6 months before I had a slight infection. I was able to clear out the infection in no time and stayed clear. I have said it more than once, but I won't use any other catheter out there, I will do everything possible to make sure I don't use any other. If Silq made this big of a change in my quality of life I can only imagine how many other lives it can and has impacted. I have spoken to other Silq users, and they feel the same way I do. I wish there was a way to let every single person that is need know that there is new technology out there other that what most facilities offer. I have even gone as far as if I see or hear of a catheter user to intervene in conversation or whatever activity they're doing at the present moment to let them know this technology is out there, and there's more than what we're being offered. In closing, I can't stop talking and sharing my experience with the Silq catheter. ***I will not be going back to any other catheter in my lifetime.***

-Dulce Garcia, 2023

On one of my mom's monthly appointments to replace the old catheter, her physician Dr Kreydin told her of a new Supra pubic product called Silq and she agreed to try it. Ever since my mom started using the new Silq Supra pubic tube, (5 months) her urinary tract infections have subsided, ***no more blockage of sediment and the tube doesn't stick to the stroma during removal for replacement of new tubes every month.*** *My mom is completely relieved and satisfied with the new Silq Supra pubic tubes. She hopes this testimony helps other patients with similar situations that may benefit from the new Silq Supra pubic tubes. It has improved my mom's health both physically and mentally. Thank you again for all you do in new and innovative methods of helping patients like my mom.*

-Maria Trevino, 2023

One day Dr. Kreydin presented the ClearTract catheter for me to use as a trial. I was immediately impressed with the difference with this catheters. There was no pain or bladder spasms on a daily basis. I noticed there were no particles floating around when irrigation was done. I do not have pain when catheter is changed. ***I also have not developed any bladder stones since using this catheter.*** *I am lucky because this catheter is being available at Rancho Los Amigos now. I do not think I would go back to the other catheters ever again. I believe it's important to make this product available to more patients who would not only benefit from the catheter but have a better quality of life with something so simple as a catheter.*

-Ana Garcia, 2023

SILQ Surface Science Perfected.

ClearTract® Clinical Evidence

Jennifer Linehan, M.D.
Associate Professor of Urology and Urologic Oncology
St. John Cancer Institute

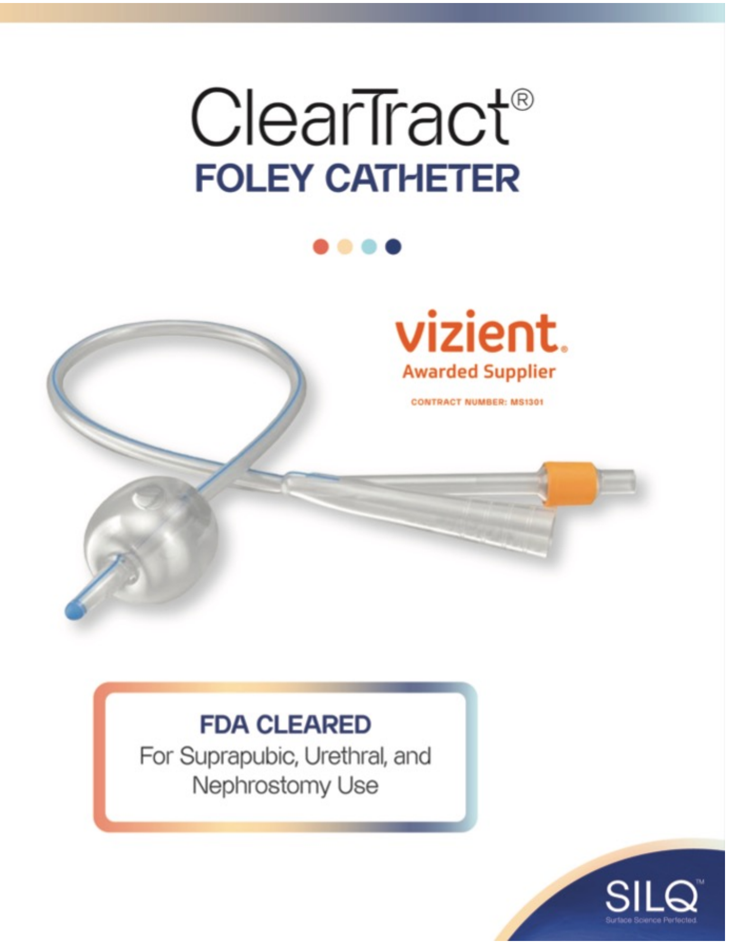
I have been using the Silq ClearTract catheters in my neurogenic bladder patients who use chronic indwelling suprapubic tubes. These are a population of patients that are at high risk for infections, colonization and the SPT often will get clogged/encrusted. Since I have been using the Silq ClearTract catheters, not only have we seen a decrease in symptomatic infection, but also decrease in obstruction and leaking from the catheter becoming clogged with sediment and debris. Just recently, I had three patients who were in the ER every other week for SPT changes because the tubes were clogged. Now I am seeing them every 4 weeks for change with no recent ER visits.

Evgeniy Kreydin, M.D.
Assistant Professor of Clinical Urology
Keck Hospital of USC
USC Norris Cancer Hospital
Verdugo Hills Hospital
Rancho Los Amigos National Rehabilitation Institute

Although urinary catheters are an integral part of any healthcare setting, they can cause significant complications, such as urinary tract infections and stone formation. In addition, urinary catheters can become obstructed or calcified. By providing a coating that resists bacterial and protein adhesion, Silq Technologies is bringing a game changing innovation to the care of patients who require catheters for bladder drainage. My early experience using these catheters indicates that they are less likely to result in infection and become obstructed, and that patients are more satisfied with their use than with the standard non-coated catheters available today.

Lora A. Plaskon, MD
Board Certified in Urology, Female Pelvic Medicine & Reconstructive Surgery
Evergreeen Health
Urogynecology Care

There have been no significant medical advances since the advent of antibiotics and handwashing that have the potential to reduce biofilm and catheter-associated infections like SILQ's zwitterion technology. Silq catheter-coating technology has the potential to revolutionize how we manage the constant threat of microbial colonization in all temporary and permanent implantable devices in humans.



ClearTract® Catheter

Brian McVerry, Ph.D.
Chief Technology Officer